

SAHAKAR SETU — Kiosk Hardware (Detailed)

PS 26088 is proposed as software + hardware. This is the hardware story: a ₹2,000–3,000 add-on turns an ordinary Android phone into a full cooperative service point.



Sahakar Setu kiosk — 3D render: Android phone (compute + display + mic), powered USB hub, amplifier + speaker, thermal receipt printer, transparent acrylic enclosure.

1. Design Principle: Why No New Hardware Is the Right Hardware Strategy

A purpose-built kiosk (Raspberry Pi 5 + touchscreen + power backup) costs **₹35,000–40,000 per unit**. India has ~1.02 lakh PACS — that would need ~₹220 crore for hardware alone (illustrative). Sahakar Setu instead repurposes a commodity Android phone (Android 8.0+, 2 GB RAM) — devices PACS staff already own or that cost ₹2,000–3,000 refurbished — and adds ₹600–900 of off-the-shelf peripherals. Same service, ~95% cheaper, and upgradable by swapping only the phone.

2. Component Breakdown (Bill of Materials)

#	Component	Spec	Role	Approx. cost
1	Android phone / tablet	Android 8.0+ (API 26+), 2 GB RAM min (4 GB preferred)	Main compute + touch display + mic + speaker fallback; runs the Capacitor kiosk APK	₹2,000–3,000 (refurbished)
2	USB OTG cable	Micro-USB or USB-C (per phone)	Phone → hub data/power bridge	₹50–100
3	Powered USB hub	4-port, externally powered	Connects printer + speaker; charges the phone; stable power for ESC/POS bursts	₹250–400
4	Amplifier circuit + speaker	3 W+ mono, 3.5 mm or USB input	Loud, clear Hindi audio for low-literacy and elderly members	₹150–250
5	Thermal receipt printer	58 mm or 80 mm, ESC/POS, USB	Prints grievance drafts, tracking IDs, receipts — the member's paper proof	₹900–1,400
6	Enclosure	Transparent acrylic / branded cardboard, phone mount, cable management	Physical presence, branding (सहकार सेतु), tamper-resistance, neat internal wiring	₹200–500
7	Power adapter	5 V / 2 A (hub + printer)	Single mains input powers the whole kiosk	₹100–150

Total per kiosk: ≈ ₹3,700–5,800 (excluding the phone) vs ₹35,000–40,000 for a Pi-based custom unit — roughly 85–90% cheaper, with the phone doubling as a backup device the PACS already has.

3. How It Fits Together

Signal / power path	Detail
Mains → 5V/2A adapter	Feeds the powered hub and the printer
Phone ⇌ hub (OTG)	Single cable; hub powers the phone and routes data
Hub → printer (USB)	ESC/POS commands from the app print drafts/receipts
Hub → amplifier → speaker	Audio out at 3 W — audible across a PACS office

Phone mic → app → server	Voice captured in-app, STT/TTS via the server (or local Vosk on-device offline mode)
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4. Software That Makes It a Kiosk (Capacitor)

The kiosk APK reuses the **same React PWA codebase** as the web client, wrapped in Capacitor with kiosk-specific native plugins:

Plugin	Why a browser can't do this
Thermal Printer	Browsers cannot reliably drive USB OTG ESC/POS printers
Kiosk Lockdown (Device Owner mode)	Member cannot exit the app — browser tabs can be closed
Auto-start on boot	Kiosk is alive after a power cut without staff intervention
Session timeout + auto-clear	No member PII left on the shared screen; forced inactivity logout
Offline cache	Cached corpus, financial-literacy audio lessons and calculators work through poor connectivity
Crash recovery	App auto-restarts; a crashed browser tab does not
Remote config	Server pushes language packs / content updates to all kiosks

Privacy rule: no member PII is ever stored on the device — everything goes straight to the server; sessions auto-clear after inactivity.

5. Power Budget

Load	Draw
Printer (print burst)	~1.5 A @ 5 V peak
Amplifier + speaker (3 W)	~0.6 A @ 5 V
Phone charging via hub	~0.5 A @ 5 V
Total (adapter 2 A)	~2.6 A peak → 5 V / 3 A adapter recommended for continuous printing

6. Optional Upgrades (future)

- Raspberry Pi 5 as alternate compute (₹35,000–40,000/kiosk only where justified).
- Larger touchscreen display where the phone screen is too small.
- Solar / battery backup for areas with unreliable power.

7. Current Status

- Kiosk APK built and installed on a Samsung A03 test device (`adb install -r`) — voice chat, grievance preview and printing paths verified live.
- Render image: `FINAL KIOSK.png` (included above) — submitted with this document.